

Station Allotment System: Comprehensive Administration Manual

1. System Overview & Architecture

The **Station Allotment System** is a robust, real-time web application engineered to manage the merit-based allocation of students to individual schools/colleges (stations) across various districts. The system handles granular complexity including overlapping academic sessions, shifting seat matrices, multi-tiered district administration, and real-time live-projected algorithm executions.

Technology Stack

- **Frontend:** React, TypeScript, Vite, Tailwind CSS, shadcn/ui.
 - **State Management:** React Query (Server State), React Hook Form.
 - **Backend Framework:** Node.js with Express.js.
 - **Database Engine:** PostgreSQL (Neon Serverless) governed by Drizzle ORM.
 - **Security:** Strict Role-Based Access Control (RBAC) separating `central_admin` and `district_admin`.
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2. Directory of Modules and Pages

The application is structured into discrete functional modules accessed via the centralized sidebar:

- **Dashboard (/):** High-level statistical overview of current rounds, total seats, allotted vs. pending students, and district participation metrics.
 - **Year Sessions (/sessions):** Infrastructure layer defining dynamic operational date ranges for academic years.
 - **Counseling Mgmt (/counseling):** The command center for creating broad titles and spawning iterative allocation rounds (Round 1 → Round 2).
 - **Vacancies (/vacancies):** Database of every available seat categorized specifically by School (UDISE), Stream, Gender, and strict Category reservations (Open, WHH, Private, Disabled).
 - **Students & Choices (/students):** Master ledger of candidate merits, their respective District, and their 1-10 preference choices.
 - **District Admins (/district-admins):** User management allowing Central to provision accounts for individual districts.
 - **District Analysis (/district-analysis):** Telemetry allowing Central to monitor how many students in each district are waiting vs. processed.
 - **Allocation Engine (/allocation):** The administrative execution layer where the merit-based matching algorithm is monitored and triggered.
 - **Projector View (/display):** A public-facing UI specifically built for large auditoriums, providing full transparency by rendering the engine's choices live.
 - **Reports & Export (/export):** Documentation hub for extracting finalized XLSX/PDF seat allotment results and cutoff statistics.
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3. Central Administration Workflow: Step-by-Step

To utilize the application successfully, a Central Administrator must follow three strict operational phases.

Phase 1: Initialization (Pre-Process Setup)

These steps must be completed before any district admin can begin interacting with students.

1. Define the Date Window:

- Navigate to **Year Sessions**.
- Create the specific variant for the season (e.g., Session **2026–2027** running from April 1 to April 10).

2. Initialize the Program:

- Navigate to **Counseling Mgmt**.
- Click **Create Title** (e.g., **Phase 1 Regular Allotment**), binding it to the session.
- The system automatically provisions **"Round 1"** in an inactive state.

3. Upload the Seat Matrix:

- Navigate to **Vacancies**.
- Import the verified CSV detailing exactly how many seats exist at each School/Category tuple.

4. Import Candidate Roster:

- Navigate to **Students**.
- Upload the absolute Merit List. Ensure everyone has an immutable **meritNumber** derived from entrance results.

5. Provision District Access:

- Navigate to **District Admins**.
- Create secure login credentials for every participating center so they can process their subset of candidates.

Phase 2: Execution & Transparency (During Process)

This is the live operational phase handling real-time data entry and algorithmic matching.

1. District Hub Processing:

- District Admins log in nationwide. They verify physical candidate documents, confirm eligibility, and input up to **10 school preferences** into the system.
- Admins exclusively use strict Action Icons (see Section 4).

2. Global Readiness Check:

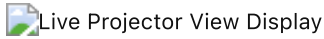
- Central Admin uses the **District Analysis** page to ensure all centers have finalized data entry.

3. Trigger the Match Engine:

- Central Admin navigates to the **Allocation** dashboard.
- Once all locks pass, the "Start Allocation" command is executed.
- *Architecture Note: The engine loads vacancies into a high-speed RAM hash map, processes students strictly ascending by merit, matches choices iteratively 1-10, and deducts seats securely.*

4. Live Auditorium Projection:

- The Central Administrator projects the **Projector View (/display)** page to monitors.
- Students in the waiting room watch the web socket dynamically render the Allocation Engine's calculations in real-time.



Direct Allocation Override / Modal Interface:

For manual interventions or highly specific edge-cases, the Administration Override Modal allows targeted allocations. These interfaces specifically guardrail choices by enforcing Category matching (Open, Disabled, Private, WHH).



Phase 3: Post-Process & Iteration (Finalization)

These steps wrap up the current block of allotments and prepare for subsequent rounds.

1. Locking the Round:

- Central Admin clicks **Finalize Allocation** . This permanently locks the snapshot. No student preferences for Round 1 can be mutated hereafter.

2. Publish formal results:

- Navigate to **Reports & Export**. Extract formal documentation containing allotments and category cutoffs.

3. Physical Admission Validation:

- As students report to their granted stations, Central/District Admins convert their status globally using the **Admit** icon.
- If a student rejects their seat, the Admin uses the **Not Admitted** icon.
- If a student leaves after admission, the Admin uses the **Vacate** icon.
- *Crucially, any Vacated or Not Admitted seat is dynamically restored to the master Vacancy pool.*

4. Spawn Subsequent Round:

- If seats remain, the Central Admin returns to **Counseling Mgmt** and clicks **Start Next Round** .
 - The system automatically snapshots the remaining vacancy pool, rolls forward unallotted candidates, and clones the setup for "Round 2".
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4. UI Operations Reference & Icons

To guarantee speed during high-traffic counseling days, student grids utilize a uniform, color-coded icon language:

Icon	Operation	Context & System Effect	Status Change
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	Unlock	Allows District Admins to explicitly re-enter the submission form if a student changes their mind prior to engine execution.	locked → pending
	Reset	Clears the student's current algorithmic allotment. Used to correct anomalies before the round is finalized.	allotted → pending
	Vacate	Explicitly drops a seat <i>after</i> it was finalized. Immediately cascades the seat back into the active network pool.	admitted/allotted → vacated
	Admit	Validates the physical arrival of the student. Final state locking the seat permanently off the market.	allotted → admitted
	Not Admitted	Flags the candidate as absent/rejected. Immediately releases the allocated seat back to the pool.	allotted → not_allotted
	Help/Flow	Toggles an on-screen flowchart explaining precisely what status is required to see these buttons.	<i>Informational</i>

Official Manual Generated for the Station Allotment Application.